COMPTE RENDU DU TP 1.1

PARTIE 1- découverte des machines actives

Machine attaquant : Kali linux (192.168.56.4)

Machine cible :

Windows 10 :192.168.56.3

Ubuntu server :192.168.56.5

```
┌──(root㉿awouna)-[/home/awouna]
└─# ping 192.168.56.4
PING 192.168.56.4 (192.168.56.4) 56(84) bytes of data.
64 bytes from 192.168.56.4: icmp_seq=1 ttl=64 time=0.223 ms
64 bytes from 192.168.56.4: icmp_seq=2 ttl=64 time=0.028 ms
64 bytes from 192.168.56.4: icmp_seq=3 ttl=64 time=0.031 ms
64 bytes from 192.168.56.4: icmp_seq=4 ttl=64 time=0.028 ms
64 bytes from 192.168.56.4: icmp_seq=5 ttl=64 time=0.060 ms
64 bytes from 192.168.56.4: icmp_seq=6 ttl=64 time=0.028 ms
^C
--- 192.168.56.4 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5105ms
rtt min/avg/max/mdev = 0.028/0.066/0.223/0.071 ms

┌──(root㉿awouna)-[/home/awouna]
└─# ping 192.168.56.5
PING 192.168.56.5 (192.168.56.5) 56(84) bytes of data.
64 bytes from 192.168.56.5: icmp_seq=1 ttl=64 time=3.12 ms
64 bytes from 192.168.56.5: icmp_seq=2 ttl=64 time=0.975 ms
64 bytes from 192.168.56.5: icmp_seq=3 ttl=64 time=1.17 ms
64 bytes from 192.168.56.5: icmp_seq=4 ttl=64 time=1.12 ms
64 bytes from 192.168.56.5: icmp_seq=5 ttl=64 time=1.00 ms
64 bytes from 192.168.56.5: icmp_seq=6 ttl=64 time=0.917 ms
64 bytes from 192.168.56.5: icmp_seq=7 ttl=64 time=1.18 ms
^C
```

1. PRÉSENTATION DE NMAP

Nmap (Network Mapper) est un outil open-source utilisé en cybersécurité et en administration réseau.

Il permet de découvrir les machines présentes sur un réseau, d'identifier les ports ouverts et les services actifs, ainsi que de détecter les systèmes d'exploitation.

Il est utilisé dans les audits de sécurité, les tests d'intrusion et le diagnostic réseau.

2. ANALYSE DES COMMANDES UTILISÉES

Option –sn

Résultat de la commande nmap –sn 192.168.56.0/24

L'image suivante montre la capture de la commande.

```
┌──(root㉿awouna)-[/home/awouna]
└─# nmap -sn 192.168.56.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-10 14:36 +0200
Nmap scan report for 192.168.56.1
Host is up (0.00059s latency).
MAC Address: 0A:00:27:00:00:0B (Unknown)
Nmap scan report for 192.168.56.2
Host is up (0.00081s latency).
MAC Address: 08:00:27:00:89:5D (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.3
Host is up (0.00087s latency).
MAC Address: 08:00:27:F1:7C:2B (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.5
Host is up (0.00095s latency).
MAC Address: 08:00:27:FA:5A:43 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.4
Host is up.
Nmap scan report for 192.168.56.30
Host is up.
Nmap done: 256 IP addresses (6 hosts up) scanned in 13.53 seconds

┌──(root㉿awouna)-[/home/awouna]
└─#
```

Permet de détecter les machines actives sans scanner les ports.

Utilisé pour un scan rapide du réseau.

Option -sS

Scan SYN (semi-ouvert), plus discret que les autres scans.

Permet de contourner certains systèmes de détection. Résultat de la commande dans l'image ci-dessus.

```
┌──(root㉿awouna)-[/home/awouna]
└─# nmap -sS 192.168.56.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 09:14 +0200
Nmap scan report for 192.168.56.5
Host is up (0.00064s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
22/tcp open  ssh
MAC Address: 08:00:27:FA:5A:43 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 5.18 seconds
```

Option -sV

Permet d'identifier les versions des services sur les ports ouverts.

Très utile pour détecter des vulnérabilités. Résultat de la commande dans l'image ci-dessus.

```
┌──(root㉿awouna)-[/home/awouna]
└─# nmap -sV 192.168.56.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 09:16 +0200
Nmap scan report for 192.168.56.5
Host is up (0.00069s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 8.2p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol
2.0)
MAC Address: 08:00:27:FA:5A:43 (Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://n
map.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.64 seconds
```

Option -O

Permet de détecter le système d'exploitation de la machine cible.

Résultat de la commande dans l'image ci-dessus nous un système linux et Windows

```
┌──(root㉿awouna)-[/home/awouna]
└─# sudo  nmap -O 192.168.56.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 09:20 +0200
Nmap scan report for 192.168.56.3
Host is up (0.0036s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
MAC Address: 08:00:27:F1:7C:2B (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Microsoft Windows 10
OS CPE: cpe:/o:microsoft:windows_10
OS details: Microsoft Windows 10 1709 - 21H2
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.o
rg/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.55 seconds
```

```
┌──(root㉿awouna)-[/home/awouna]
└─# sudo  nmap -O 192.168.56.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 09:19 +0200
Nmap scan report for 192.168.56.5
Host is up (0.0014s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
22/tcp open  ssh
MAC Address: 08:00:27:FA:5A:43 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4)
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.o
rg/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.50 seconds
```

Option -A

Scan complet incluant détection OS, services, scripts et traceroute. . Résultat de la commande dans l'image ci-dessus.

```
┌──(root㉿awouna)-[/home/awouna]
└─# nmap -A 192.168.56.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 09:26 +0200
Nmap scan report for 192.168.56.5
Host is up (0.0011s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 8.2p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol
2.0)
| ssh-hostkey:
|   3072 4f:e1:2b:13:07:4c:3c:79:ad:cf:43:3a:a8:84:0f:43 (RSA)
|   256 f2:74:80:a8:2a:93:e7:0e:2c:d3:b7:3c:08:e5:50:84 (ECDSA)
|_  256 98:76:dc:29:57:3c:b7:33:6d:7a:71:8f:98:a7:ac:ee (ED25519)
MAC Address: 08:00:27:FA:5A:43 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4)
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT     ADDRESS
1   1.14 ms 192.168.56.5

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.31 seconds
```

3. RÉSULTATS OBTENUS

Les machines détectées incluent les ordinateurs (win10 et Ubuntu-server 20)

Les ports ouverts courants sont :

-pour Windows 10

ports	statut	services
135/tcp	open	msrpc
139/tcp	open	Netbios-ssn
445/tcp	open	Microsoft-ds(SMB)

-pour Ubuntu-server

22/tcp	open	ssh

Les services identifiés permettent de comprendre les applications actives sur les machines.

4. ANALYSE DE SÉCURITÉ

Certains services comme SMB et SSH sont considérés comme dangereux car ils ne sont pas sécurisés.

Les ports inutilisés doivent être fermés pour réduire la surface d'attaque.

Les risques incluent l'accès non autorisé, le vol de données et les attaques informatiques.

5. UTILISATION AVANCÉE DE NMAP

Option -p-

Permet de scanner tous les ports (1 à 65535). Dans les images suivantes on peut voir les ports ouvert présent dans les deux machines ainsi que leur service.

```
┌──(root㉿awouna)-[/home/awouna]
└─# nmap -p- 192.168.56.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 09:45 +0200
Nmap scan report for 192.168.56.5
Host is up (0.00066s latency).
Not shown: 65534 closed tcp ports (reset)
PORT    STATE SERVICE
22/tcp open  ssh
MAC Address: 08:00:27:FA:5A:43 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 40.68 seconds
```

```
┌──(root㉿awouna)-[/home/awouna]
└─# nmap -p- 192.168.56.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 09:50 +0200
Nmap scan report for 192.168.56.3
Host is up (0.0021s latency).
Not shown: 65524 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
5040/tcp  open  unknown
49664/tcp open  unknown
49665/tcp open  unknown
49666/tcp open  unknown
49667/tcp open  unknown
49668/tcp open  unknown
49669/tcp open  unknown
49670/tcp open  unknown
MAC Address: 08:00:27:F1:7C:2B (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 59.56 seconds
```

Option --script vuln

Permet de détecter automatiquement les vulnérabilités connues. Image de la commande ci-dessus nous montre que le service de partage de fichier est accessible mais vulnérable sur la machine Windows, car le système bloque les tentatives de connexion.

```
┌──(root㉿awouna)-[/home/awouna]
└─# nmap -script vuln 192.168.56.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 09:54 +0200
Nmap scan report for 192.168.56.3
Host is up (0.00062s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
MAC Address: 08:00:27:F1:7C:2B (Oracle VirtualBox virtual NIC)

Host script results:
|_samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: Failed to receive bytes: ERR
R
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_smb-vuln-ms10-054: false

Nmap done: 1 IP address (1 host up) scanned in 29.49 seconds
```

Sur la machine Ubuntu aucune vulnérabilité malgré le port ssh soit ouvert l'image suivante nous montre le resultat.

```
┌──(root㉿awouna)-[/home/awouna]
└─# nmap -script vuln 192.168.56.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 10:06 +0200
Stats: 0:00:10 elapsed; 0 hosts completed (0 up), 1 undergoing ARP Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Nmap scan report for 192.168.56.5
Host is up (0.00059s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
22/tcp open  ssh
MAC Address: 08:00:27:FA:5A:43 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 15.36 seconds
```

Option -T4

Permet d'accélérer la vitesse du scan. A T4 la vitesse augmente on peut passer de 15s a 6s. L'image suivante nous montre le résultat.

```
┌──(root㉿awouna)-[/home/awouna]
└─# nmap -T4 -A 192.168.56.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-15 10:27 +0200
Nmap scan report for 192.168.56.5
Host is up (0.0014s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 8.2p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   3072 4f:e1:2b:13:07:4c:3c:79:ad:cf:43:3a:a8:84:0f:43 (RSA)
|   256 f2:74:80:a8:2a:93:e7:0e:2c:d3:b7:3c:08:e5:50:84 (ECDSA)
|_  256 98:76:dc:29:57:3c:b7:33:6d:7a:71:8f:98:a7:ac:ee (ED25519)
MAC Address: 08:00:27:FA:5A:43 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4)
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT     ADDRESS
1   1.37 ms 192.168.56.5

OS and Service detection performed. Please report any incorrect results at https://nmap.org/sub
mit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.99 seconds
```

Option –Sv –Sc -0

Scanne avec détection d'os et version des protocoles.

```
┌──(root㉿awouna)-[/home/awouna]
└─# nmap -sv -sC -0 192.168.56.3
Nmap 7.98 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
```

```
  -b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
    Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
  --exclude-ports <port ranges>: Exclude the specified ports from scanning
  -F: Fast mode - Scan fewer ports than the default scan
  -r: Scan ports sequentially - don't randomize
  --top-ports <number>: Scan <number> most common ports
  --port-ratio <ratio>: Scan ports more common than <ratio>
SERVICE/VERSION DETECTION:
  -sV: Probe open ports to determine service/version info
  --version-intensity <level>: Set from 0 (light) to 9 (try all probes)
  --version-light: Limit to most likely probes (intensity 2)
  --version-all: Try every single probe (intensity 9)
  --version-trace: Show detailed version scan activity (for debugging)
SCRIPT SCAN:
  -sC: equivalent to --script=default
  --script=<Lua scripts>: <Lua scripts> is a comma separated list of
           directories, script-files or script-categories
  --script-args=<n1=v1,[n2=v2,...]>: provide arguments to scripts
  --script-args-file=filename: provide NSE script args in a file
  --script-trace: Show all data sent and received
  --script-updatedb: Update the script database.
  --script-help=<Lua scripts>: Show help about scripts.
           <Lua scripts> is a comma-separated list of script-files or
           script-categories.
OS DETECTION:
  -O: Enable OS detection
  --osscan-limit: Limit OS detection to promising targets
  --osscan-guess: Guess OS more aggressively
TIMING AND PERFORMANCE:
```

CONCLUSION

Nmap est un outil puissant pour analyser et sécuriser un réseau. Son utilisation doit être légale et éthique.